

MA388 Sabermetrics: Lesson 19

Comparing Peak Trajectories - Part II

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Comparing Trajectories of Home Run Hitters

1. Find the ten players in baseball history who have hit the most career home runs.

```
library(tidyverse)
library(Lahman)
library(knitr)
```

```
Top10 <- Batting |>
  summarize(totalHR = sum(HR), .by = 'playerID') |>
  slice_max(totalHR, n=10) |>
  inner_join(People, by = "playerID") |>
  mutate(label = str_c(nameFirst, nameLast, sep=' ')) |>
  relocate(label, totalHR)
```

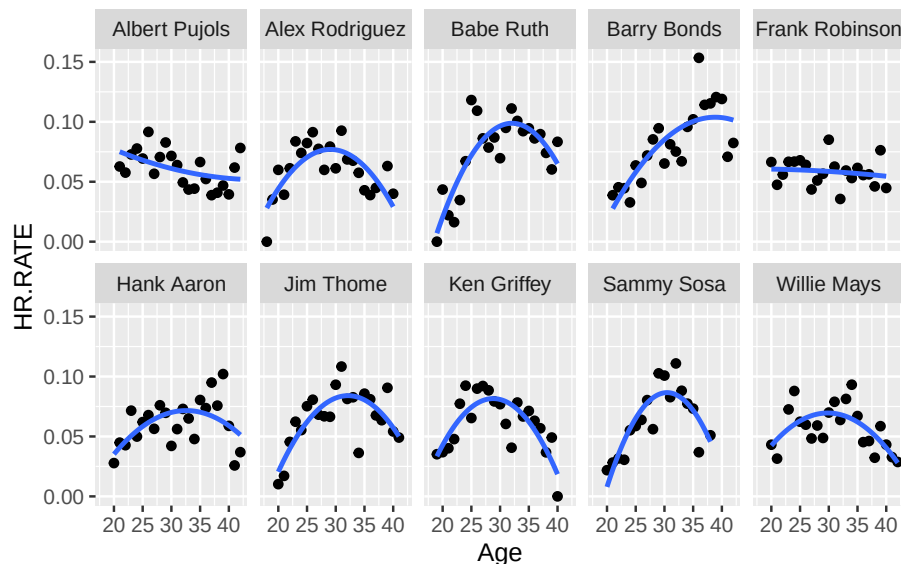
```
Top10 |>
  select(label, totalHR, debut) |>
  kable()
```

label	totalHR	debut
Barry Bonds	762	1986-05-30
Hank Aaron	755	1954-04-13
Babe Ruth	714	1914-07-11
Albert Pujols	703	2001-04-02
Alex Rodriguez	696	1994-07-08
Willie Mays	660	1951-05-25
Ken Griffey	630	1989-04-03
Jim Thome	612	1991-09-04
Sammy Sosa	609	1989-06-16
Frank Robinson	586	1956-04-17

2. Fit quadratic functions to the home run rates of the ten players, where $HR.RATE = HR / AB$. Display the fitted trajectories in a single figure.

```
top10_stats <- Batting |>
  filter(playerID %in% Top10$playerID) |>
  summarize(HR.RATE = sum(HR)/sum(AB), .by=c('playerID', 'yearID')) |>
  inner_join(People, by = "playerID") |>
  mutate(birthyear = ifelse(birthMonth >= 7,
                           birthYear + 1, birthYear),
         label = str_c(nameFirst, nameLast, sep=' '),
         Age = yearID - birthyear)

top10_stats |>
  ggplot(aes(x = Age, y = HR.RATE)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y~poly(x,2), se = FALSE) +
  facet_wrap(~label, ncol = 5)
```



3. On the basis of your work, which player had the highest estimated home run rate at his peak? Which player among the ten had the smallest peak home run rate?

```
top10_model <- function(this_playerID){
  this_player_stats <- top10_stats |>
```

```

    filter(playerID == this_playerID)

    fit <- lm(HR.RATE ~ I(Age - 30) + I((Age - 30)^2), data = this_player_stats)

    A <- coef(fit)[1]
    B <- coef(fit)[2]
    C <- coef(fit)[3]

    # Return all the stuff you just obtained.
    tibble(label = head(this_player_stats$label,1), playerID = this_playerID, A, B, C,
           age.max = 30 - B / C / 2,
           max = A - B^2 / C / 4)
  }

# Map the model function to the list of playerIDs you obtained in 1.
HR_results <- map_df(Top10$playerID, top10_model)
HR_results |>
  kable()

```

label	play- erID	A	B	C	age.max	max
Barry Bonds	bondsba01	0.0851324	0.0042375	-	38.82412	0.1038285
Hank Aaron	aaronha01	0.0700492	0.0011897	-	32.59352	0.0715920
Babe Ruth	ruthba01	0.0965013	0.0022230	-	32.06727	0.0987990
Albert Pujols	pujo- lal01	0.0610317	-	0.0000387	45.60271	0.0515987
Alex Rodriguez	ro- drial01	0.0766162	-	-	29.06393	0.0769664
Willie Mays	mayswi01	0.0695671	-	-	29.83585	0.0695747
Ken Griffey	grif- fke02	0.0808547	-	-	28.79225	0.0815873
Jim Thome	thomeji01	0.0820880	0.0018386	-	32.12711	0.0840435
Sammy Sosa	sosasa01	0.0864312	0.0006562	-	30.45580	0.0865808
Frank Robinson	robinfr02	0.0586096	-	-	15.61690	0.0607582

```
# Highest estimated max HR rate:
```

```
HR_results |>  
  slice_max(max)
```

```
# A tibble: 1 x 7
```

	label	playerID	A	B	C	age.max	max
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Barry Bonds	bondsba01	0.0851	0.00424	-0.000240	38.8	0.104

```
# Lowest estimated max HR rate:
```

```
HR_results |>  
  slice_min(max)
```

```
# A tibble: 1 x 7
```

	label	playerID	A	B	C	age.max	max
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Albert Pujols	pujolal01	0.0610	-0.00121	0.0000387	45.6	0.0516

Your answer here.

4. Do any of the players have unusual career trajectory shapes? Is there any possible explanation for these unusual shapes?

Your answer here.